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Electrolyte-gated transistors based on phenyl-C61-butyric acid methyl ester (PCBM) films: bridging redox properties, charge carrier transport and device performance

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The n-type organic semiconductor phenyl-C61-butyric acid methyl ester (PCBM), a soluble fullerene derivative well investigated for organic solar cells and transistors, can undergo several successive reversible, diffusion-controlled, one-electron reduction processes. We exploited such processes to shed light on the correlation between electron transfer properties, ionic and electronic transport as well as device performance in ionic liquid (IL)-gated transistors. Two ILs were considered, based on bis(trifluoromethylsulfonyl)imide [TFSI] as the anion and 1-ethyl-3-methylimidazolium [EMIM] or 1-butyl-1-methylpyrrolidinium [PYR_{14}] as the cation. The aromatic structure of EMIM and its lower hindrance with respect to [PYR_{14}] favor a 3D (bulk) electrochemical doping. As opposed to that, for PYR_{14} the doping seems to be 2D (surface-confined). If the n-doping of the PCBM is pursued beyond the first electrochemical process, the transistor current vs gate-source voltage plots in [PYR_{14}][TFSI] feature a maximum that points to the presence of finite windows of high conductivity in IL-gated PCBM transistors.

Electrolyte-gated transistors (EGTs) exploit electrical double layers, forming at the interface between an ion-including medium and a semiconductor, to modulate the electrical conductance in the transistor channel.[1] The high capacitance (typically 10~40 μF cm^{-2}) of thin electrical double layers (ca 2-4 nm-thick) permits to reach high charge carrier density in semiconducting channels (ca 10^{15} cm^{-2}), at low operating voltages (as low as 0.1-0.5 V). EGTs based on channels of 2D materials, metal oxides and organic molecular semiconductors, making use of different gating media, such as ionic liquids (ILs), ion gels and saline aqueous solutions, have been reported in the literature. Among such channel materials, organic semiconductors are interesting for applications in bioelectronics as well as flexible and printable electronics.[2-4] The working principle of EGTs depends on several factors, such as the nature of the transistor channel material and the gating medium. It is reasonable to hypothesize that the softness and the redox activity featured by organic semiconductors render them prone to a bulk (3 dimensional, 3D) electrochemical doping, where charge carrier injection/extraction from metal electrodes (source and drain) to the semiconductor is assisted by the incorporation/removal of ions in the channel. Nevertheless, specific contributions pertaining to intermolecular interactions between the ions of the gating medium and the organic semiconducting molecules of the transistors channel, as well as the size of the ions, can dramatically affect the effectiveness of the doping and its extension in the film (3D vs 2D). Therefore, such specific contributions have to be considered to properly describe the operation of organic EGTs.

Organic semiconductors based on fullerenes and their derivatives have been employed in organic bulk heterojunction solar cells and organic transistors.[5-8] Fullerene-based materials have been widely investigated in the past decades for their electrochemical properties, both in solution and thin film form.[9-10] C_{60} in tetrahydrofurane (THF) solutions, with supporting electrolyte tetrabutylammonium hexafluorophosphate $(C_4H_9)_4NPF_6$, at -60 °C, can undergo six successive reversible, diffusion-controlled, one-electron reduction processes, whereas the same number of reductions had been previously observed only in solvent mixtures.[11-12] Cyclic voltammetry has been recently used, in parallel with low-temperature photoelectron spectroscopy, to estimate the location of the lowest unoccupied molecular orbital (LUMO) levels of phenyl-C61-butyric acid methyl ester (PCBM) in film form, a fullerene derivative soluble in common organic solvents and, as such, interesting for solution-based thin film technologies. Three quasi reversible reduction processes were observed in the solvent o-dichlorobenzene including the support electrolyte tetrabutylammonium tetrafluoroborate $(TBA)BF_4$.[13] Interestingly, literature reports that the electrochemical behavior of C_{60} in thin film form is more complex with respect to solution counterparts.[9] The nature of the supporting electrolyte cation affects the reduction processes, in terms of structure, free energy and solubility of the C_{60} phases formed by electrochemical doping. In general, it has been observed that the larger the doping cation, the larger is the peak splitting in the cyclic voltammograms. Scanning Electrochemical Microscopy (SECM) experiments conducted in

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acetonitrile containing $(TBA)BF_4$ revealed that the partial reduction of the films is paralleled by an increase in their conductivity. Considering that PCBM can undergo several redox processes, we found particularly intriguing to explore, in IL-gated transistor configuration, possible correlations between the redox state of the PCBM, the nature of the ions of the IL gating medium and the charge carrier transport and device properties of the PCBM transistor channel material. Our interest was fortified by the possibility to observe the presence of a finite window of high conductivity upon reducing electrochemically the PCBM films beyond the first reduction process. Such windows of high conductivity have been already reported in the literature for p-type organic polymer semiconductors.[14]

In this work, we report on PCBM transistors gated by two ILs based on bis(trifluoromethylsulfonyl)imide [TFSI] as the anion and 1-ethyl-3-methylimidazolium [EMIM] or 1-butyl-1-methylpyrrolidinium $[PYR_{14}]$ as the cation. We used atomic force microscopy (AFM) and X-ray Diffraction (XRD) to gain insight on the structure of the PCBM films. We studied the film charge transfer properties by cyclic voltammetry. Afterwards, IL-gated PCBM transistors were characterized at different sweep rates and different intervals of the gate-source voltage, to gain insight on the interplay of ionic and electronic transport and the effect of the advancement of the electrochemical doping on charge transport.

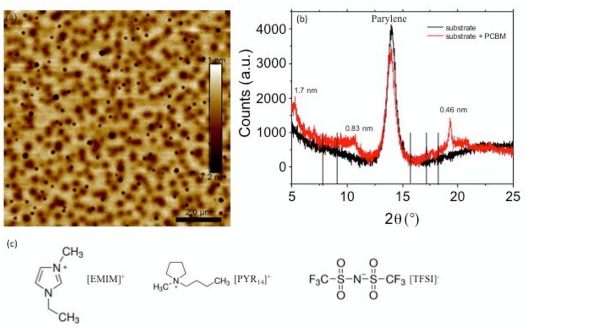

Fig. 1 (a) 10 μm×10 μm AFM image of PCBM films deposited on SiO_2, height scale: -7.2 nm to 4.1 nm; (b) XRD patterns of the PCBM films including parylene wells confining the ionic liquid gating medium (see ESI).[18]

AFM images show that PCBM films spin-coated on SiO_2 completely cover the substrate and have root mean square roughness (rms) of ca 2.0 nm (Fig. 1a). The XRD patterns of the films (Fig. 1b) show 3 peaks located at 5.3° (corresponding to an interplanar distance of 1.7 nm), 10.7° (0.83 nm) and 19.3° (0.46 nm), in agreement with the XRD patterns of polycrystalline PCBM films.[15] After characterization, we exposed the films to the ILs [EMIM][TFSI] and $[PYR_{14}]$[TFSI] to study their charge transfer properties in these media, by cyclic voltammetry. [EMIM][TFSI] and $[PYR_{14}]$[TFSI] feature ionic conductivities of 8.8 mS cm^{-1} and 2.6 mS cm^{-1}, viscosities of 36 mPa s and 84 mPa s at room temperature and wide electrochemical stability windows (4.1 V and 5.7 V).[16] The cyclic voltammetry experiments were carried out in transistor configuration. Here, the PCBM film included between source and drain electrodes constituted the working electrode and a high surface area activated carbon gate electrode acted as both the counter and the reference electrode.[17] Within the interval of the electrochemical potential included between 0.5 V and -1.9 V vs carbon reference, we observed several redox processes in the cyclic voltammograms (Fig. 2). In [EMIM][TFSI], PCBM electrodes featured 3 reduction processes during the forward (cathodic) scan. Using a sweep rate of 100 mV·s^{-1} (Fig. 2a), the cathodic peaks (for the 3rd cycle) were located at ca. -1 V (shoulder), -1.4 V (peak) and -1.65 V (peak). Three corresponding anodic peaks, located at ca. -1.45 V, -1.1 V and -0.6 V together with a broad shoulder at ca. -0.2 V, were observed

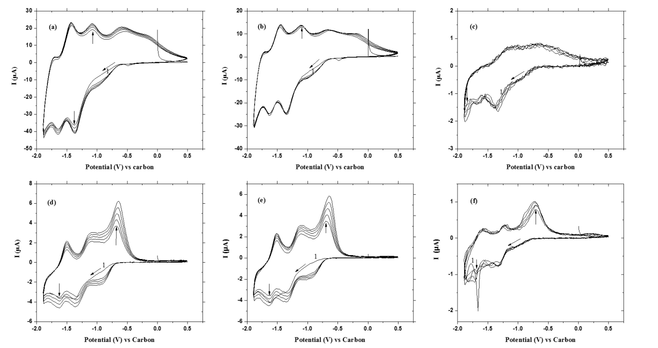

Fig. 2 Cyclic voltammograms obtained in transistor configuration for PCBM films in [EMIM][TFSI] (a, b, c) and $[PYR_{14}]$[TFSI] (d, e, f), with sweep rates of 100 mV·s^{-1} (a, d), 50 mV·s^{-1}(b, e) and 1 mV·s^{-1} (c, f). The quasi reference electrode is made of high surface area activated carbon. Only the first five cycles are shown.

in the backward (anodic) scan. In $[PYR_{14}]$[TFSI] (Fig. 2d), the 3 reduction processes have cathodic peaks located at ca. -1 V (shoulder), -1.4 V (peak) and -1.65 V (peak) whereas 3 corresponding anodic peaks were observed, in the backward scan, at ca. -1.5 V, -1.05 V and -0.7 V. The small and broad shoulder at -0.2 V, reported for [EMIM][TFSI], was completely absent in $[PYR_{14}]$[TFSI], thus suggesting that such a shoulder can be attributed to the re-oxidation of some kind of reduced form of PCBM product, most probably identified as a partially protonated reduced PCBM, considering the acidic protons belonging to [EMIM]. It is worth noticing that the oxidation peak at -0.7 V in $[PYR_{14}]$[TFSI] has a *triangular* shape, typical of surface redox processes. The hindrance of $[PYR_{14}]$ with respect to [EMIM] could lead to the significant presence of surface-confined redox processes in $[PYR_{14}]$[TFSI] with respect to [EMIM][TFSI]. Theoretical calculations, assuming a spherical shape for the ions, permitted an estimation of the volume of the cations of the ILs: 182 Å^3 for [EMIM] (leading to a radius of

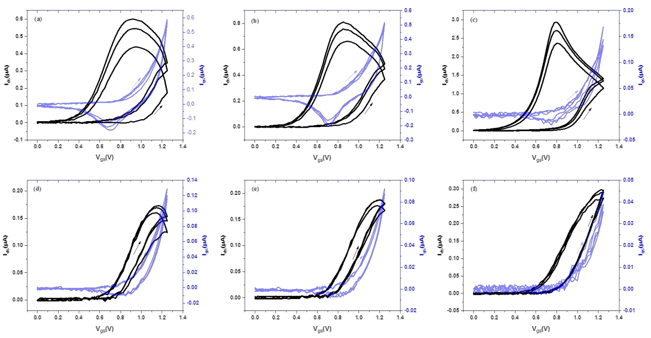

Fig. 3 Transfer characteristics of PCBM transistors making use of (a, b, c) [EMIM][TFSI] and (d, e, f) $[PYR_{14}]$[TFSI] as the gating medium, obtained for V_{gs} between 0 V and 1.25 V at different sweep rates: (a, d) 100 mV·s^{-1}, (b, e) 50 mV·s^{-1},(c, f) 1 mV·s^{-1}. V_{ds}=0.1 V.

ca. 3.5 Å) and 253 Å^3 for $[PYR_{14}]$ (radius ca. 3.9 Å).[19] Different sweep rates were explored to gain insight on the electron transfer processes in the two media. The voltammetric currents are, for a fixed rate, higher in [EMIM][TFSI] than $[PYR_{14}]$[TFSI], likely due to the better ionic conductivity of [EMIM][TFSI]. Other contributions could also explain the higher current in [EMIM][TFSI], e.g. the lower steric hindrance of [EMIM] (and its acidic proton) vs [PYR_{14}], which translates into an easier access/removal of the ion to the reduction/oxidation of PCBM. The decrease of the sweep rate led to minor changes in the peak positions and a decrease in the current (Table S1).[20,21]

The knowledge gained about the electrochemical behavior of the PCBM films constituted the underpinning to design electrolyte-gated transistors using [EMIM][TFSI] and [PYR_{14}][TFSI] as the gating media. The use of the high surface area activated carbon gate enabled low voltage (sub-1 V) transistor operation and permitted to avoid the use of an external reference electrode, thus simplifying the device structure.[17] We initially characterized transistors for values of V_{gs} included between 0 V and 1.25 V. The devices showed *n*-type transistor behavior and worked in accumulation mode (Fig. 3, Fig. S2). It is apparent from the transfer characteristics that the threshold voltages of the transistors have values of about 0.8 V. The linear transfer characteristics of PCBM transistors, at different sweep rates, show that the decrease of the rate leads to higher transistor current (I_{ds}). At lower sweep rates, doping ions better accommodate in the channel. At 1 mV s^{-1}, I_{ds} is ca. 4 times higher than at 100 mV·s^{-1} for [EMIM][TFSI] whereas it is slightly less than 2 times higher for [PYR_{14}][TFSI]. A higher I_{ds} is observed in [EMIM][TFSI]-gated transistors compared to [PYR_{14}][TFSI], indicating a higher doping effectiveness with [EMIM] compared to [PYR_{14}], in agreement with the cyclic voltammetry. The higher I_{ds}, together with the significant hysteresis observed with [EMIM][TFSI], points to a 3D (bulk) doping mechanism for PCBM in [EMIM][TFSI]. The doping in [PYR_{14}][TFSI] is still electrochemical but it likely involves only the surface of the PCBM film (surface-confined doping, two dimensional, 2D). The difference in the doping mechanisms can be attributed to the different steric hindrance of the two IL cations (higher for [PYR_{14}]) but also to their different molecular structure (the aromatic [EMIM] is expected to have a higher affinity with PCBM with respect to [PYR_{14}], i.e. to be more easily intercalated in the PCBM films to give a 3D doping, instead of a 2D one). The presence of the acidic proton in [EMIM] should also be considered, for [EMIM]. The aromatic nature of [EMIM] can establish a cooperative and strong π-π stacking interactions with the reduced caps of the fullerene, leading to a higher number of counterions in comparison to the case of [PYR_{14}].[22] The *thinner* structure of [EMIM], in comparison to that of [PYR_{14}], permits [EMIM] to slide easily in the PCBM network. The charge carrier density, *p*, in our PCBM channels was deduced from the equation $p = \frac{Q}{eA} = \frac{\int I_{ds} dV_{gs}}{r_v eA}$ where *Q* is the accumulated charge during the forward scan in the transfer curve (obtained through the integration of I_{gs} with V_{gs}), *A* is the geometric area of the PCBM film interfaced to the electrolyte, *e* is the elementary charge and r_v is the sweep rate. The charge carrier densities we obtained at sweep rates of 1, 50 and 100 mV·s^{-1} were 3×10^{15}, 2×10^{14} and 10^{14} cm^{-2} in [EMIM][TFSI] and 8×10^{14}, 3×10^{13} and 2×10^{13} cm^{-2} in [PYR_{14}][TFSI]. The electron mobility, μ, was obtained through $\mu = \frac{L}{W}\frac{I_{ds}}{V_{dsep}}$ where *L* is the source and drain interelectrode distance, 10 µm, and *W* is the width, 4 mm. The values of the mobility for transistors making use of [EMIM][TFSI] at sweep rates of 1, 50 and 100 mV s^{-1} were ca. 7×10^{-6}, 3×10^{-4} and 7×10^{-4} cm^2 V^{-1} s^{-1} whereas they were 8×10^{-6}, 4×10^{-4} and 8×10^{-4} cm^2 V^{-1} s^{-1} in [PYR_{14}][TFSI]. The values of the mobility are similar for the 2 gating media, but lower with respect to PCBM transistors making use of SiO_2 or organic gate dielectrics.[23] The relatively low mobility is attributable to the disruption of the $\pi-\pi$ packing taking place at different extents, depending on the nature of the doping mechanism (a more pronounced disruption being expected for the bulk doping). On the other hand, the values of the mobility are strongly affected by the sweep rate of V_{gs}. We observe that, increasing the sweeping rate from 1 mV s^{-1} to 50 mV s^{-1}, the charge density accumulated in the PCBM films decreases of roughly one order of magnitude, whereas the mobility increases of about two orders of magnitude. We could deduce that high sweep rates lead to lower charge carrier density, in turn leading to high charge carrier mobility. Looking carefully at the results, we can also see that the charge carrier densities, for the same sweep rate, are lower for [PYR_{14}][TFSI] than for [EMIM][TFSI], although the values of the mobility are similar. This deduction seems to point to the 2D vs 3D doping observable in the two cases, respectively. The transistor ON(1 V)/OFF(0 V) ratio was about 70-100, for both [EMIM][TFSI] and [PYR_{14}][TFSI]-gated transistors.

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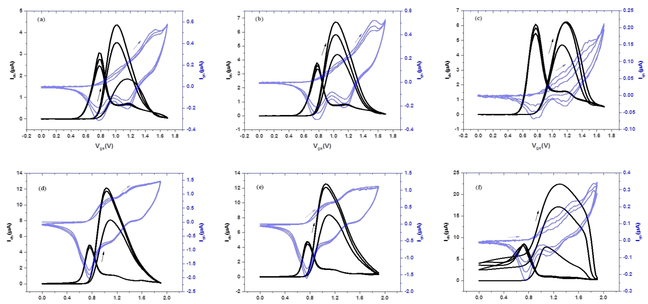

Fig. 4 Transfer characteristics of PCBM transistors making use of [PYR_{14}][TFSI] as the gating medium, at different V_{gs} sweep rates: (a, d) 100 mVs^{-1}, (b, e) 50 mVs^{-1}, (c, f) 1 mVs^{-1}. The range of V_{gs} values (compare with Fig. 3) permits to study the behavior of the transistors beyond the first electrochemical doping process. For (a), (b) and (c) the interval of V_{gs} is included between 0 V-1.7 V; for (d), (e) and (f), between 0 V-1.9 V.

We later extended the characterization of the transistors to higher values of V_{gs}, namely 1.7 V and 1.9 V (Fig. 4 and Fig. S3). For the [PYR14][TFSI] case, Fig. 4 shows dramatic differences with respect to when V_{gs} limited to 1.25 V (Fig. 3). In [PYR_{14}][TFSI], for V_{gs} up to 1.7 V, during the first I_{ds} vs V_{gs} forward scan, one peak is observable during the 1st cycle (3 cycles are shown in Fig. 4), located at ca 1.2 V for 100 mV s^{-1}, 1.1 V for 50 mV s^{-1} and ca 1.2 V at 1 mV s^{-1}. The peaks at ca 1.2 V and 1.1 V are more intense during the 2nd and 3rd cycle, with respect the first. There is a shift of the peak position from the 1st to the last 2 cycles. In the backward scan, two peaks are observable; the peak positions do not change significantly from the 1st to the last 2 cycles. The decrease of the rate from 100 to 50 and 1 mV s^{-1} leads, as expected, to an increase of I_{ds}. I_{gs} vs V_{gs} plots recorded at 100, 50 and 1 mV s^{-1} show, during the forward scan of the 1st cycle, one shoulder at ca 1.1 V and a peak at ca 1.5 V, and, on the backward scan, 3 peaks at high sweep rates and only 2 peaks at 1 mV s^{-1}, in agreement with the voltammetric results. The shape of the curves is similar for V_{gs} up to 1.9 V, apart from the case of 1 mV s^{-1}, where the effect of the 3rd injection in the PCBM seems observable in the transfer plots. We propose that the kinetics of the

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3rd injection is affected by the interaction with the IL cations themselves, thus making the process more sluggish than the 1st or the 2nd ones, as it is observable only at 1 mV s^{-1}. The maximum in the transfer curves when the doping takes place beyond the 1st redox process ($V_{gs} \geq 1.25$ V), is analogous to that of p-type polymers featuring finite windows of high conductivity.[14] Band filling and bipolaron formation (fullerene C60 itself forms bipolarons at sufficiently negative potentials[24]) at high densities can explain the decrease in conductivity.

In conclusion, to gain insight into possible correlations between the electrochemical and the charge carrier transport properties of organic semiconductors, we investigated the doping process in n-type PCBM transistors making use of [EMIM][TFSI] and [PYR_{14}][TFSI] as the gating media and high surface area carbon as the gate electrode. Cyclic voltammetry permitted to observe 3 reduction processes in the films, thus opening the possibility to follow the film doping for different reduction states. Our transistor characterization points to a predominant 2D, surface-confined electrochemical doping with the [PYR_{14}][TFSI] and a predominant 3D (bulk) doping with [EMIM][TFSI]. Preliminary results obtained using 4 M [Li][TFSI] in tetraethylene glycol dimethyl-ether (TEGDME) as the gating medium ("solvent in salt", radius of the solvated Li ion ca. 2.5 Å) confirm our hypothesis on the dramatic effect of the steric hindrance of the cation on the doping mode of the PCBM films.[25,26,27] Indeed, the characteristics of the transistors making use of 4 M [Li][TFSI] in TEGDME resemble those observed with [EMIM][TFSI], interpreted as bulk doping (Fig. S4 and Fig. S5). Interestingly, in [PYR_{14}][TFSI], the shape of I_{ds} reminds that of p-type organic semiconductors featuring finite windows of high conductivity, thus suggesting that these windows are a general feature of organic semiconductors.

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